

Grade F

OCTOLAN NETWORK ASSESSMENT

Riverside Dental Group (sample data)

Friday 30 October 2026, 07:58 · scan time 26m0s · collector TECH-LAPTOP-07 (windows/amd64)

OctoLAN identified 36 devices (36 responding during the assessment) across 2 subnet(s), including 5 network infrastructure devices. No critical issues were found, but 5 high-severity issues should be remediated promptly. 4 device(s) run software past end of support.

36	5	2	3
DEVICES	INFRASTRUCTURE	SUBNETS	VLANS

0	5	6	4
CRITICAL	HIGH	MEDIUM	END-OF-LIFE

Priority actions

Severity	Issue	Affected
HIGH	**End-of-life operating system** Plan upgrade or replacement; isolate on a restricted VLAN in the meantime.	4 device(s)
HIGH	**SNMP responds to a default community string** Change community strings to long random values, restrict SNMP by ACL to the monitoring host, or move to SNMPv3 authPriv.	2 device(s)
HIGH	**Telnet enabled** Disable Telnet and use SSH. On Cisco: `line vty 0 15` → `transport input ssh`.	2 device(s)
HIGH	**SMBv1 enabled** Disable SMB1: `Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol`; on NAS devices set minimum protocol to SMB2.	1 device(s)
HIGH	**Multiple DHCP servers on one segment** Identify the unauthorised server; enable DHCP snooping on access switches.	network
MEDIUM	**Management web UI over plain HTTP** Enable HTTPS-only management and disable the HTTP listener.	7 device(s)

Network overview

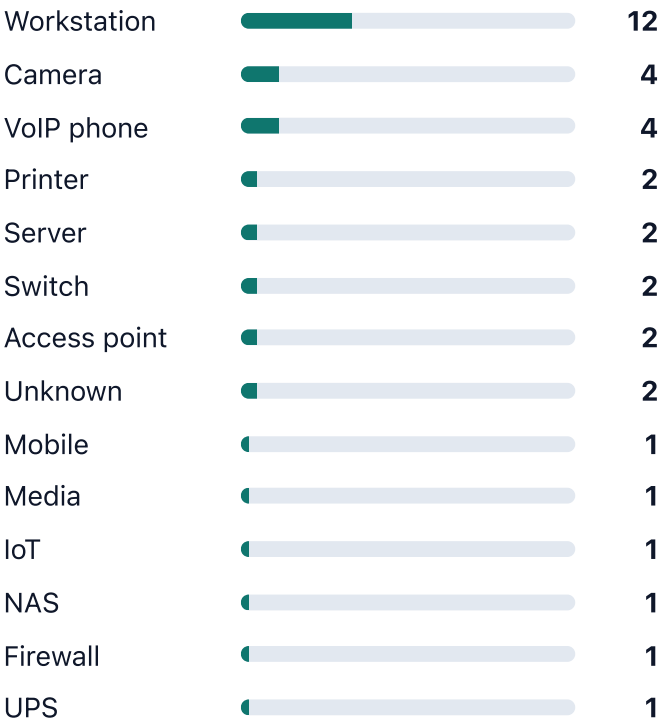
Subnets

10.20.0.0/24	collector segment · gateway 10.20.0.1
10.20.10.0/24	scanned · gateway 10.20.10.1

DHCP servers

10.20.0.10	router 10.20.0.1 · DNS 10.20.0.10 · riverside.local
192.168.0.1	router 192.168.0.1 · DNS 192.168.0.1

Device mix

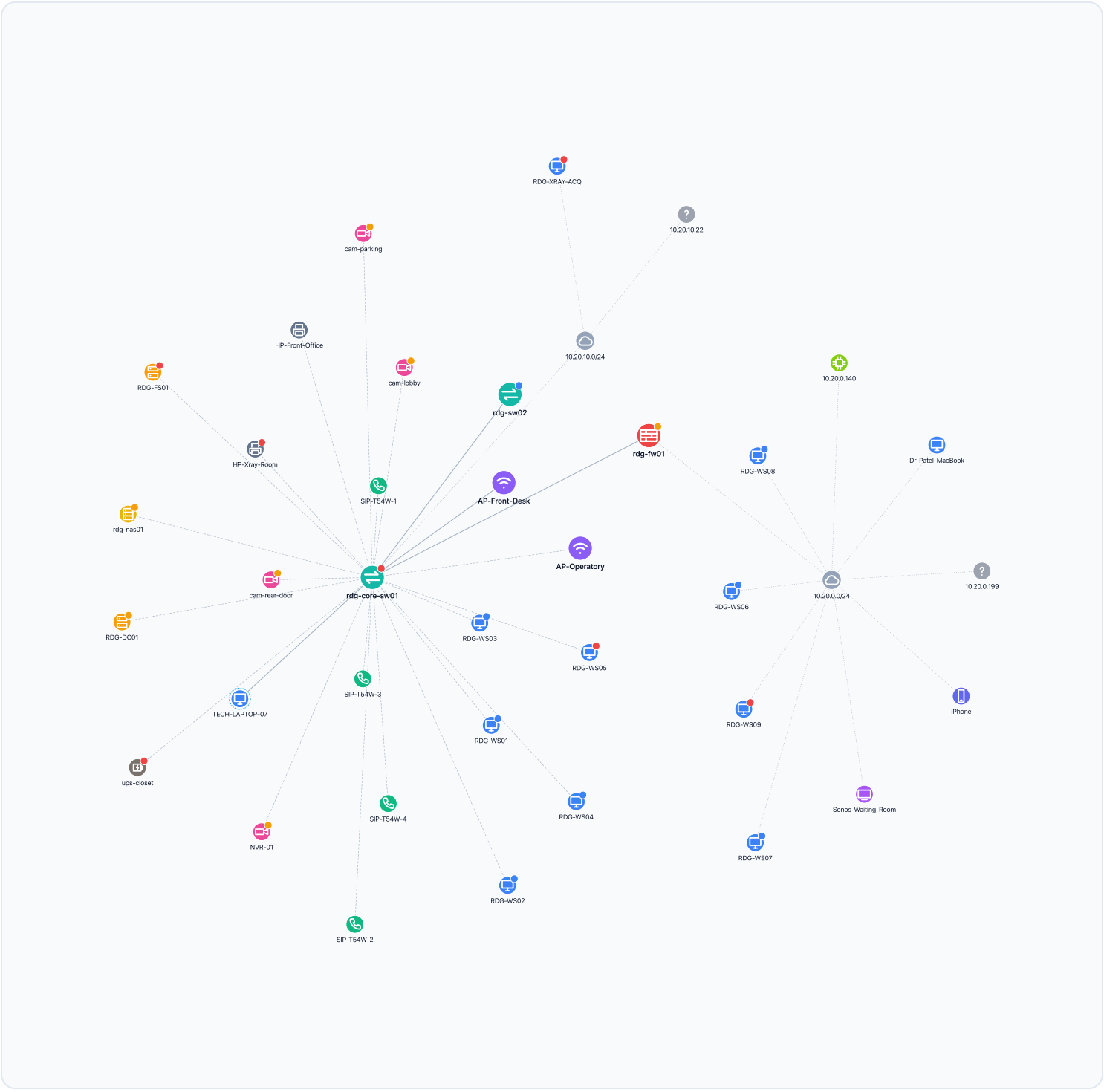


Layer 2

RSTP (802.1w) root bridge 32769.00:00:0c:4a:21:20, 4 topology change(s) observed.
Peak broadcast+multicast: **2750** bcast / **340** mcast pps.

Topology

Solid lines are LLDP/CDP adjacencies, dashed lines come from switch MAC tables, dotted lines are inferred from subnet membership. Drag to pan, scroll to zoom.



Findings

End-of-life operating system

Lifecycle HIGH

These devices run an OS or firmware past vendor end of support and no longer receive security patches.

Recommendation: Plan upgrade or replacement; isolate on a restricted VLAN in the meantime.

- RDG-FS01 (10.20.0.15)
- RDG-WS05 (10.20.0.104)
- RDG-WS09 (10.20.0.108)
- RDG-XRAY-ACQ (10.20.10.21)

SNMP responds to a default community string

Credentials HIGH

The device answers SNMP with 'public' or 'private'. Anyone on the LAN can read its configuration details; 'private' may allow writes.

Recommendation: Change community strings to long random values, restrict SNMP by ACL to the monitoring host, or move to SNMPv3 authPriv.

- rdg-core-sw01 (10.20.0.2)
- ups-closet (10.20.0.25)

Telnet enabled

Insecure protocols HIGH

Telnet sends credentials and sessions in cleartext. Anyone on the network path can capture admin passwords.

Recommendation: Disable Telnet and use SSH. On Cisco: `line vty 0 15` → `transport input ssh`.

- rdg-core-sw01 (10.20.0.2)
- HP-Xray-Room (10.20.0.31)

SMBv1 enabled

Vulnerable services HIGH

SMBv1 is the protocol abused by WannaCry/NotPetya (EternalBlue) and is deprecated by Microsoft.

Recommendation: Disable SMB1: `Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol`; on NAS devices set minimum protocol to SMB2.

- RDG-FS01 (10.20.0.15)

Multiple DHCP servers on one segment

Network HIGH

More than one DHCP server answered: 10.20.0.10, 192.168.0.1. A rogue DHCP server can redirect clients' gateway/DNS (man-in-the-middle) or cause outages.

Recommendation: Identify the unauthorised server; enable DHCP snooping on access switches.

Management web UI over plain HTTP

Insecure protocols MEDIUM

Infrastructure devices offer their admin page without HTTPS, exposing admin credentials on the wire.

Recommendation: Enable HTTPS-only management and disable the HTTP listener.

- rdg-nas01 (10.20.0.20)
- ups-closet (10.20.0.25)
- HP-Xray-Room (10.20.0.31)
- NVR-01 (10.20.0.40)
- cam-lobby (10.20.0.41)
- cam-parking (10.20.0.42)
- cam-rear-door (10.20.0.43)

SMB signing not required on servers

Hardening MEDIUM

Without required signing, SMB relay attacks (e.g. NTLM relay) can impersonate users against these servers.

Recommendation: Set 'Microsoft network server: Digitally sign communications (always)' via GPO.

- RDG-DC01 (10.20.0.10)
- RDG-FS01 (10.20.0.15)

Expired TLS certificates

Certificates MEDIUM

Services are presenting certificates that have already expired; users learn to click through warnings.

Recommendation: Renew the certificates (or automate with ACME/internal CA).

- rdg-fw01 (10.20.0.1)

FTP service exposed

Insecure protocols MEDIUM

FTP transfers credentials and files unencrypted.

Recommendation: Replace with SFTP/FTPS, or disable the service if it is not needed.

- RDG-FS01 (10.20.0.15)

Switch/router interfaces with errors

Performance MEDIUM

Interfaces report a high error count (CRC/frame errors), commonly a bad cable, duplex mismatch or failing optic.

Recommendation: Check cabling and duplex settings on the listed devices' ports.

- rdg-core-sw01 (10.20.0.2)

UPnP enabled on the gateway

Hardening MEDIUM

UPnP lets any internal device open inbound ports on the firewall without admin approval.

Recommendation: Disable UPnP/NAT-PMP on the edge device.

- rdg-fw01 (10.20.0.1)

RDP enabled on workstations

Remote access LOW

Remote Desktop listening on endpoints widens the lateral-movement surface for ransomware.

Recommendation: Disable RDP on workstations or restrict it by firewall to admin jump hosts; require NLA.

- RDG-WS01 (10.20.0.100)
- RDG-WS02 (10.20.0.101)
- RDG-WS03 (10.20.0.102)
- RDG-WS04 (10.20.0.103)
- RDG-WS05 (10.20.0.104)
- RDG-WS06 (10.20.0.105)
- RDG-WS07 (10.20.0.106)
- RDG-WS08 (10.20.0.107)
- RDG-WS09 (10.20.0.108)
- RDG-XRAY-ACQ (10.20.10.21)

SNMP v1/v2c in use

Insecure protocols LOW

Community strings travel in cleartext.

Recommendation: Migrate monitoring to SNMPv3 with authentication and privacy (SHA/AES).

- rdg-fw01 (10.20.0.1)
- rdg-core-sw01 (10.20.0.2)
- rdg-sw02 (10.20.0.3)
- ups-closet (10.20.0.25)

Infrastructure not rebooted in over a year

Lifecycle LOW

Uptime over a year usually means firmware has not been patched.

Recommendation: Schedule firmware updates in a maintenance window.

- rdg-core-sw01 (10.20.0.2)

TLS certificates expiring within 30 days

Certificates LOW

Certificates will expire soon and cause outages or browser warnings.

Recommendation: Renew before expiry.

- rdg-nas01 (10.20.0.20)

Self-signed certificates on management interfaces

Certificates INFO

Self-signed certificates can't be distinguished from an attacker's certificate by users.

Recommendation: Issue certificates from an internal CA and distribute the root via GPO/MDM.

- rdg-fw01 (10.20.0.1)
- rdg-nas01 (10.20.0.20)

Unidentified devices

Inventory INFO

These devices answered on the network but OctoLAN could not determine what they are (no hostname, open services, SNMP or vendor match).

Recommendation: Identify and label each device. Unknown devices are a common source of shadow IT and unmanaged risk.

- 10.20.0.199
- 10.20.10.22

Device inventory

Name	IP	MAC / vendor	Type	Operating system	Open TCP ports	Latency
rdg-fw01 FortiGate-60F S/N FGT60FTK2209XXXX	10.20.0.1	00:09:0f:aa:10:01 Fortinet	Firewall	FortiGate-60F v7.2.8,build1639	443 8443 22	410 µs
rdg-core-sw01 WS-C2960X-48FPD-L S/N F0C2138XXXX	10.20.0.2	00:00:0c:4a:21:10 Cisco	Switch	Cisco IOS Software, C2960X Software (C2960X-UNIVERSALK9-M), Version 15.2(7)E4	22 23 80 443	417 µs
rdg-sw02 C1000-24P-4G-L	10.20.0.3	00:00:0c:4a:21:20 Cisco	Switch	Cisco IOS Software, C1000 Software (C1000-UNIVERSALK9-M), Version 15.2(7)E8	22	594 µs
RDG-DC01	10.20.0.10	00:06:5b:10:00:10 Dell	Server	Windows Server 2022 (build 20348)	53 88 135 139 389 445 3389	475 µs
AP-Front-Desk	10.20.0.11	00:15:6d:31:00:01 Ubiquiti	Access point	Linux/Unix	22	1.3 ms

AP-Operatory	10.20.0.12	00:15:6d:31:00:02 Ubiquiti	Access point	Linux/Unix	22	1.5 ms
RDG-FS01	10.20.0.15	00:06:5b:10:00:15 Dell	Server	Windows Server 2012 R2 (build 9600) EOL	135 139 445 3389 21	552 µs
rdg-nas01	10.20.0.20	00:11:32:aa:bb:20 Synology	NAS	Linux/Unix	22 80 139 445 5000 5001	869 µs
ups-closet	10.20.0.25	00:c0:b7:55:00:25 American Power Conversion	UPS	APC Web/SNMP Management Card (MB:v4.1.0 PF:v6.4.6 PN:apc_hw05_aos_646.bin)	80	1.2 ms
HP-Front-Office HP LaserJet Pro M404dn	10.20.0.30	00:01:e6:30:00:30 Hewlett Packard	Printer	Linux/Unix	80 443 631 9100	2.1 ms
HP-Xray-Room	10.20.0.31	00:01:e6:30:00:31 Hewlett Packard	Printer	Linux/Unix	80 9100 23	2.5 ms
NVR-01	10.20.0.40	00:bc:99:40:00:40 Hangzhou Hikvision Digital	Camera	Linux/Unix	80 554 8000	1.1 ms
cam-lobby	10.20.0.41	00:bc:99:40:00:41 Hangzhou Hikvision Digital	Camera	Linux/Unix	80 554	1.3 ms
cam-parking	10.20.0.42	00:bc:99:40:00:42 Hangzhou Hikvision Digital	Camera	Linux/Unix	80 554	1.3 ms
cam-rear-door	10.20.0.43	00:bc:99:40:00:43 Hangzhou Hikvision Digital	Camera	Linux/Unix	80 554	1.4 ms
TECH-LAPTOP-07	10.20.0.57	00:12:fe:99:00:57 Lenovo	Workstation	Windows 11 24H2		
SIP-T54W-1	10.20.0.60	24:9a:d8:00:00:60 Yealink	VoIP phone	Linux/Unix	80 443 5060	965 µs
SIP-T54W-2	10.20.0.61	24:9a:d8:00:00:61 Yealink	VoIP phone	Linux/Unix	80 443 5060	972 µs

SIP-T54W-3	10.20.0.62	24:9a:d8:00:00:62 Yealink	VoIP phone	Linux/Unix	80 443 5060	1.1 ms
SIP-T54W-4	10.20.0.63	24:9a:d8:00:00:63 Yealink	VoIP phone	Linux/Unix	80 443 5060	1.9 ms
RDG-WS01	10.20.0.100	00:12:fe:00:01:00 Lenovo	Workstation	Windows 11 24H2 (build 26100)	135 139 445 3389	2.1 ms
RDG-WS02	10.20.0.101	00:12:fe:00:01:01 Lenovo	Workstation	Windows 11 24H2 (build 26100)	135 139 445 3389	1.0 ms
RDG-WS03	10.20.0.102	00:12:fe:00:01:02 Lenovo	Workstation	Windows 11 23H2 (build 22631)	135 139 445 3389	1.1 ms
RDG-WS04	10.20.0.103	00:12:fe:00:01:03 Lenovo	Workstation	Windows 11 24H2 (build 26100)	135 139 445 3389	1.4 ms
RDG-WS05	10.20.0.104	00:12:fe:00:01:04 Lenovo	Workstation	Windows 10 22H2 (build 19045) **EOL**	135 139 445 3389	1.8 ms
RDG-WS06	10.20.0.105	00:12:fe:00:01:05 Lenovo	Workstation	Windows 11 24H2 (build 26100)	135 139 445 3389	2.7 ms
RDG-WS07	10.20.0.106	00:12:fe:00:01:06 Lenovo	Workstation	Windows 11 23H2 (build 22631)	135 139 445 3389	3.0 ms
RDG-WS08	10.20.0.107	00:12:fe:00:01:07 Lenovo	Workstation	Windows 11 24H2 (build 26100)	135 139 445 3389	1.7 ms
RDG-WS09	10.20.0.108	00:12:fe:00:01:08 Lenovo	Workstation	Windows 10 22H2 (build 19045) **EOL**	135 139 445 3389	1.8 ms
Dr-Patel-MacBook Mac15,6	10.20.0.120	00:03:93:00:01:20 Apple	Workstation	macOS (Mac15,6)		1.8 ms

Sonos-Waiting-Room	10.20.0.130	00:0e:58:00:01:30 Sonos	Media	Linux/Unix		1.9 ms
10.20.0.140	10.20.0.140	00:4b:12:00:01:40 Espressif	IoT	Linux/Unix	80	1.5 ms
iPhone	10.20.0.150	da:a1:19:00:01:50 private MAC	Mobile	Apple iOS/iPadOS	62078	1.5 ms
10.20.0.199	10.20.0.199		Unknown	Linux/Unix		1.7 ms
RDG-XRAY-ACQ	10.20.10.21		Workstation	Windows 7 (build 7601) EOL	135 139 445 3389	2.4 ms
10.20.10.22	10.20.10.22		Unknown	Linux/Unix	104 80	2.6 ms

Traffic (NetFlow / IPFIX)

Top talkers

10.20.0.15	16.8 GiB
10.20.0.40	13.0 GiB
10.20.0.20	8.5 GiB
10.20.0.105	3.0 GiB
52.96.164.18	2.7 GiB

Top applications

SMB	19.6 GiB
HTTPS	15.8 GiB
RTSP	11.2 GiB
RDP	2.0 GiB
DNS	381.5 MiB

Methodology

The assessment combined active discovery (ARP and ICMP sweeps, TCP service checks, SNMP polling, SMB/NTLM and TLS inspection, reverse DNS, NetBIOS and mDNS) with passive listening (DHCP, LLDP/CDP, mDNS/SSDP, spanning tree and broadcast metering). No credentials were used other than the SNMP communities supplied, and no exploits or intrusive tests were run.

Phase	Result	Detail
Prepare	done	Ethernet (10.20.0.57) · 508 targets (254 local, 254 routed)
ARP sweep	done	38 hosts via raw ARP frames
Ping sweep	done	31 replies (windows-api ICMP)
TCP discovery	done	4 additional hosts found

Service scan	done	141 open ports on 39 hosts
Name resolution	done	Named 41/41 hosts
SNMP polling	done	5 SNMP agents polled
Service discovery	done	Queried mDNS/Bonjour and SSDP/UPnP
Passive listening	done	Listened for DHCP, LLDP/CDP, STP
Analysis	done	

Collector capabilities: raw capture available; ICMP mode windows-api.